OO-E-990c
April 10, 1967

SUPERSEDING
Int. Fed. Spec. 00-E-00990b (Army-GL)
September 9, 1966 and
Fed. Spec. 00-E-990a
July 26, 1962

FEDERAL SPECIFICATION

EXTRACTOR, LAUNDRY, COMMERCIAL

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers commercial laundry extractors.

1.1.1 *Federal specification coverage.* Federal specifications do not include all varieties of commodity as indicated by the title of the specifications, or which are commercially available, but are intended to cover only those generally used by the Federal Government.

1.2 Classification.

1.2.1 *Types and sizes.* Commercial laundry extractors covered by this specification shall be of the following types and sizes as specified (see 6.2):

Type I—Rigid-curb type.
- Size 20—20 inch, 25 pound capacity.
- Size 24—24 inch, 50 pound capacity.
- Size 30—30 inch, 70 pound capacity.

Type II—Open top type.
- Size 40—40 inch, 150 pound capacity.
- Size 48—48 inch, 250 pound capacity.

Type III—Unloading type.
- Size 54—54 inch, 375 pound capacity.
- Size 60—60 inch, 450 pound capacity.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

CC-M-636—Motor, Alternating-Current, (Fractional Horsepower).
CC-M-641—Motor, Alternating Current, (Integral Horsepower, 200 HP and Smaller).
QQ-B-671—Bronze Castings, Aluminum.
QQ-N-281—Nickel-Copper-Alloy Bar, Plate, Rod, Sheet, Strip, Wire, Forgings, and Structural and Special Shaped Sections.
QQ-N-288—Nickel-Copper Alloy and Nickel-Copper-Silicon Alloy Castings.

Federal Standards:

Fed. Std. No. 66—Steel: Chemical Composition and Hardenability.
Fed. Std. No. 123—Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 595—Colors.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo, Ft. Worth, Denver, San Francisco, Los Angeles, and Auburn, Wash.

(Federal Government activities may obtain copies

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of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Specifications:

MIL-T-152—Treatment, Moisture and Fungus Resistant, of Communications, Electronic, and Associated Electrical Equipment.

MIL-V-173—Varnish, Moisture and Fungus Resistant, for the Treatment of Communications, Electronic, and Associated Electrical Equipment.

MIL-L-3153—Laundry and Dry Cleaning Machinery and Equipment (for Fixed Installations), Preparation for Delivery for Domestic and Overseas Shipment.

MIL-I-16910—Interference Measurement, Radio, Methods and Limits; 14 Kilocycles to 1000 Megacycles.

MIL-E-55301—Electromagnetic Compatibility.

Military Standards:

MIL-STD-105—Sampling Procedures and Tables for Inspection by Attributes.

MIL-STD-129—Marking for Shipment and Storage.

MIL-STD-130—Identification Marking of U.S. Military Property.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issues in effect on date of invitation for bids or request for proposal shall apply.

Underwriters' Laboratories, (UL) Inc. Standard:

No. 508—Industrial Control Equipment.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 1285 Walt Whitman Road, Melville, Long Island, N. Y., 11746; 207 East Ohio St., Chicago, Ill., 60611; or 1655 Scott Blvd., Santa Clara, Calif., 95050.)

National Fire Protection Association (NFPA) Standard:

No. 70—National Electrical Code.

(Application for copies should be addressed to the National Fire Protection Association, 60 Batterymarch St., Boston, Mass., 02110.)

American Welding Society (AWS) Code:

Standard Qualification Procedures of the American Welding Society.

(Application for copies should be addressed to the American Welding Society, United Engineering Center, 345 E. 47th St., New York, N.Y., 10017.)

American Society of Mechanical Engineers (ASME) Code:

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 E. 47th St., New York, N. Y., 10017.)

National Bureau of Standards (NBS) Handbook:

H28—Screw Thread Standards for Federal Services.

(Application for copies should be addressed to the Superintendent of Documents U. S. Government Printing Office, Washington, D. C., 20402.)

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

3. REQUIREMENTS

3.1 Preproduction sample. When specified (see 6.2), before production is commenced, a sample finished extractor shall be submitted or made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the preproduction sample authorizes the commencement of production, but does not relieve the contractor of responsibility for compliance with all other applicable provisions of this specification. The preproduction sample shall be manufactured in the

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same facilities to be used for the manufacture of the production items.

3.2 **Standard product.** The extractors furnished under this specification shall be the manufacturer's current standard product except for those modifications or additional changes required in that standard product in order to comply with the requirements of this specification. All accessories and components shall be furnished with each extractor that are normally furnished with the standard product offered commercially.

3.3 **Fire and casualty hazards.**

3.3.1 *Conformance to UL, Inc.* The contractor shall submit to the contracting agency, proof that the electrical components of the extractor he proposes to supply under this specification, except for motors (see 3.5.2.1.2.1), conform to the requirements of the UL, Inc., as applicable. Any electrical motor controls shall conform to UL Standard No. 508. The label, or listing with re-examination, of the UL, Inc. may be accepted as evidence that the electrical components of the extractor conform to this requirement. In lieu of the label, or listing with re-examination, the contractor may submit independent proof, satisfactory to the contracting agency, that the electrical components conform to the published standards including methods of tests of the UL, Inc.

3.3.1.1 *Contractor compliance.* Compliance with UL, Inc. requirements regarding fire and casualty hazards does not absolve the contractor from complete compliance with the requirements of this specification in order to secure acceptance of his extractor.

3.3.2 *Safety standard.* Electrical wiring and equipment of laundry extractors shall conform to the requirements of NFPA Standard No. 70.

3.3.2.1 *Certification.* Acceptable evidence of meeting the requirements of NFPA Standards shall be the contractor's certified statement that the extractor conforms to the applicable requirements. Compliance with NFPA regarding safety does not absolve the contractor from compliance with the requirements of this specification in order to secure acceptance of his extractor.

3.4 **Materials.** The materials used shall conform to the applicable specifications and requirements specified hereinafter. When materials are not definitely specified, they shall be of the quality normally used for the purpose in commercial practice.

3.4.1 *Corrosion-resisting metals.*

3.4.1.1 *Corrosion-resisting steel.* The corrosion-resisting steel shall conform to AISI numbers 302, 304 or 316 of Fed. Std. No. 66.

3.4.1.2 *Nickel-copper alloy.* The nickel-copper alloy shall conform to QQ-N-281 or QQ-N-288.

3.4.1.3 *Bronze aluminum castings.* The bronze aluminum castings shall conform to QQ-B-671.

3.4.2 *Materials for fasteners.* Rivets, bolts, screws, nuts, and washers shall be of steel, except where brass or corrosion-resisting metal is to be fastened, in which case they shall be of brass or corrosion-resisting metal, respectively. Where dissimilar metals are fastened, rivets, bolts, screws, nuts, and washers shall be of corrosion-resisting metal. Electrodes or welding rods used for welding shall deposit material similar to the base metal. All screw threads shall comply with NBS H28. Unified screw threads shall be used in lieu of American National screw threads and nonunified American Standard screw threads.

3.5 **Design and Construction.**

3.5.1 *Design.* The extractors shall be centrifugally operated and conform to the following designs:

Type I—Rigid-curb type (nonflexibly supported), and basket which shall not have removable containers.

Type II—Open-top type, shall have the

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curb flexibly supported from the base or the basket flexibly supported from the curb and the basket shall not have removable containers.

Type III—Unloading type shall have a curb flexibly supported from the base and a permanent basket, with removable containers, supported from the curb.

3.5.2 *General construction.* Extractors shall be furnished complete with motors, driving mechanism, starters, and controllers, including master switches, timers, and other electrical equipment and commercial accessories, as applicable. Construction shall conform to the general and detail requirements specified hereinafter. The extractors shall be tested as specified in 4.4.1 and 4.4.2 to determine conformance to the following requirements:

(a) The assembly of components shall have complete and proper inter-relationship.
(b) There shall be no leak in the curb assembly. Slight seepage of water from the cover will be permitted.
(c) Moving parts and bearings shall evidence no overheating.
(d) Starting and stopping devices shall operate smoothly and positively.
(e) There shall be no structural failures.

3.5.2.1 *Electrical system and equipment.*

3.5.2.1.1 *Electrical characteristics.* The electrical components of the extractors shall be suitable for the voltage, frequency, and number of phases for AC operation or the voltage for DC operation, as specified (see 6.2).

3.5.2.1.2 *Motors and controls.*

3.5.2.1.2.1 *Motors.* The motors shall be drip proof or totally enclosed type, and capable of operating in a 40°C. ambient temperature. Motors shall conform to the requirements of CC-M-636 or CC-M-641, as applicable, and to the following requirements.

(a) All motors shall have windings impregnated to resist moisture.
(b) Motors shall be equipped with ball bearings, except that motors of less than 1/2 horsepower for horizontal application may have sleeve bearings.
(c) Motor bearings shall be of the permanently lubricated type or shall have adequate and accessible means for lubrication.
(d) Each motor shall be provided with a manually reset inverse-time-thermal-overload protection incorporated in the controller.

3.5.2.1.2.2 *Motors and starters.* Motors and starters shall be suitable for starting 40 inch size and larger extractors once every 10 minutes, and smaller size extractors once every 6 minutes without exceeding the allowable temperature rise under full load when in operation for not less than one hour. Conformance to these requirements shall be determined when tested as specified in 4.4.2. The motor for driving the extractor shall be a type suitable for mounting the shaft in a vertical position. The motor shall be rigidly mounted in an adjustable manner on a suitable support attached to the curb or secured to the base of the extractor. The motor shall not be mounted under the extractor basket, except for machines smaller than 30-inch diameter.

3.5.2.1.2.3 *Motor controller circuit.* Each extractor shall be provided with a controller for each motor timer, limit switch, remote control pushbutton switch, signal and indicating light, as necessary to perform the intended functions. The controllers shall be of the magnetic across-the-line type having a separate pole for each ungrounded conductor and shall conform to the requirements of UL Standard No. 508. When direct-current motors are specified (see 3.5.2.1.1) and are of more than 1/2 horsepower, the controllers shall be of the reduced-voltage type having one or more steps of armature resistance

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for smooth acceleration. The starters or controllers shall be of the contactor type for mounting on the machine, the wall, or a stand adjacent to the extractor and shall be operated by enclosed pushbuttons located on the curb, curb support, or stand. Pushbutton control stations shall be mounted at points convenient to the operator(s), and shall be fully protected or recessed to prevent accidental operation.

3.5.2.1.3 *Conduit and wire, electric.* Electrical conduit shall be rigid or flexible steel or electrical metallic tubing. All electrical wire shall be heat resistant grade, thermoplastic insulated.

3.5.2.1.4 *Wiring and instructions.* Wiring shall be provided between all components mounted on the extractor using materials conforming to materials specified in 3.5.2.1.3, as applicable, except flexible metallic conduit may be used where required for belt adjustment by moving a motor. Installation shall conform to the requirements of the NFPA Standard No. 70. All wiring shall terminate in a connection box with provisions for connections beyond the machine. Wiring and connection diagrams including instructions for connecting controllers to be mounted separately from the machine shall be furnished with the machine. Wiring beyond the machine will be furnished by the purchaser.

3.5.2.2 *Extractor bases.* Extractor bases shall be constructed so that points of contact with the floor shall be in the same plane. Provisions shall be made for anchoring the machine to the floor with sufficient anchoring to prevent undue vibration and movement of the machine during full load operation. Conformance to this requirement shall be determined when the equipment is tested as specified in 4.4.2. Channels, I-beams, anchor bolts, and other foundation materials, except concrete, and a diagram of recommended foundation details shall be furnished with each extractor.

3.5.2.3 *Spacing and extent of fastenings.* The spacing and extent of welds, rivets, bolts, and screws shall be such as to insure proper fastening and prevent bulging of the metals fastened.

3.5.2.4 *Drive.* The extractor basket shall be driven by V-belt(s). Belts shall be protected from water.

3.5.2.5 *Speed warnings.* When specified (see 6.2), baskets larger than 24 inches shall have the rated maximum speed of the basket in revolutions per minute (r.p.m.) indented or embossed on the inside of the basket, in readily visible letters and figures not less than 1/4 inch high. The rated maximum speed shall be stamped on the nameplate. The direction of basket rotation shall be indicated and readily visible.

3.5.2.6 *Tolerances.* Unless otherwise specified herein, tolerances shall be in accordance with the manufacturer's commercial practice.

3.5.2.7 *Lubrication.* Means shall be provided to insure proper lubrication of the spindle bearing and other moving parts. Means shall also be provided to prevent the lubricant from coming in contact with the items being extracted or the rubber rings, if used (see 4.4.2).

3.5.2.7.1 *Lubrication instructions.* Lubrication instructions including location, type of lubricant, and time intervals shall be indented or embossed on a noncorrodible metal plate, permanently affixed in a convenient and conspicuous location.

3.5.2.8 *Radio interference suppression.*

3.5.2.8.1 *Army purchases.* Equipment procured under this specification for Army use shall be designed and equipped for conformance to the administrative requirements of MIL-E-55301 (see 4.4.3.1).

3.5.2.8.2 *Navy and Air Force purchases.* When specified (see 6.2), equipment procured under this specification for Navy and Air Force use shall conform to MIL-I-16910 relative to radiated and conducted radio in-

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terference. Radio interference suppression and tests therefore will not be required for electric motors without commutation having not more than one starting contact and operated at 3600 or less r.p.m. (see 4.4.3.2).

3.5.2.9 *Finish and color.* Unless otherwise specified (see 6.2), extractors (except for unpainted components), shall be finished according to the manufacturer's standard practice. Unless otherwise specified (see 6.2), the color of the components to be painted shall correspond to No. 24172 of Fed. Std. No. 595. Emergency stop and normal stop pushbuttons shall be red and this color shall be limited to stop pushbuttons.

3.5.2.9.1 *Metallic coating thickness.* When galvanizing or other protective metallic coating is used, the thickness of the coating shall be in accordance with the manufacturer's standard commercial practice.

3.5.2.9.2 *Finish of exposed corrosion-resisting metals.* Corrosion-resisting steel and nickel-copper alloy metal shall be free from scale. Exposed corrosion-resisting metal surfaces shall be finished to a commercial polish.

3.5.2.10 *Manuals, instruction and parts.*

3.5.2.10.1 *Instruction books.* Unless otherwise specified (see 6.2), one instruction book giving operating, installation, and maintenance instructions (including pictures or illustrations as necessary), dimensions, floor plans of machine, electrical data, and connection diagrams for all utilities, shall be furnished and packed with each extractor.

3.5.2.10.2 *Parts manual.* Unless otherwise specified (see 6.2), one copy of the manufacturer's illustrated repair parts manual shall be furnished with each extractor. The manual shall include the supplier's part number, the part number assigned by the actual manufacturer, name of actual manufacturer, and description of the parts. Each specific repair part shall be assigned a specific part number, and part numbers will not be duplicated by a given manufacturer regardless of model number or end item application, unless an interchangeable factor exists; which in this case, the manufacturer will assign identical part numbers to identical parts. Figure, drawing, item, plate, casting, or numbers may be used for cross reference only and will not be used in lieu of part numbers. The requirements specified in this paragraph may be included in the instruction books specified in 3.5.2.10.1.

3.5.2.11 *Name and data plates.* Except on procurements for civil agencies, each extractor and replacement part covered by this specification shall be marked for identification in accordance with MIL-STD-130. For procurements for civil agencies, the plates may be in accordance with the manufacturer's standard practice provided they are legibly and permanently marked with the manufacturer's name or trademark of such known character that the source of manufacture of the extractor may be readily determined, and contain the type or model and size of the machine, and the electrical characteristics, as applicable.

3.5.2.12 *Moisture and fungus resistant treatment.* When specified (see 6.2), electrical components of equipment covered by this specification shall be moisture and fungus resistant treated with materials conforming to and applied as required by MIL-V-173 and MIL-T-152, as applicable.

3.5.3 *Construction details.*

3.5.3.1 *Basket diameters, speed and capacities.* The baskets shall accelerate and decelerate smoothly and shall have the diameters, capacities, and speeds specified in table I (see 4.4.2).

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TABLE I. *Basket diameter, capacity and speed*

Type	Inside diameter of basket, minimum	Capacity per load, dry weight, minimum	Revolutions per minute of basket, minimum
	Inches	*Pounds*	*R.P.M.*
Rigid curb	20	25	1,500
	24	50	1,200
	30	70	1,000
Open top	40	150	850
	48	250	750
Unloading	54	375	750
	60	450	700

TABLE II. *Extractor drainage connections*

Size of extractor	Size of individual outlets (minimum)
Inches	*Inches*
20	1-1/2
24	1-1/2
30	3
40	4
48	4
54	4
60	4

3.5.3.2 *Curb.* The curb shall be of corrosion-resisting metal not less than 0.0335 inch thick for the 20-, 24-, and 30-inch extractors. It shall be designed to permit ready access to the balancing and driving mechanism, and shall be provided with a covered inspection port to this mechanism. The curb top and side shall be fabricated from corrosion resisting metal not less than 0.085 inch thick for the 40- and 48- inch diameter sizes, and not less than 0.100 inch thick for the 54- and 60-inch diameter sizes. Each curb top and side shall be properly secured and sealed to a ferrous metal base. In lieu of the corrosion resisting metal curbs specified for 20-, 24-, and 30-inch extractors, the extractor may be supplied with a cast ferrous metal curb rigidly secured to a one piece ferrous metal base. The cast curb assembly shall be rigidly secured to the machine base or the curb and machine base may be a single casting. All joints in the curb shall be watertight.

3.5.3.2.1 *Forming curb top.* Where corrosion-resisting metal is required, the top of the curb shall be fitted with a corrosion-resisting metal ring set close to the basket, or other means shall be provided to prevent water splashing from the extractor, and permit ready removal of the basket as a unit (see 4.4.2).

3.5.3.2.2 *Drainage connections.* A drainage connection or connections shall be fitted at the lowest point of the curb and shall be of the minimum sizes specified in table II.

3.5.3.3 *Baskets.* The side sheet of the baskets shall be of corrosion-resisting metal not less than 0.100 inch thick for extractor baskets over 30 inches in diameter, not less than 0.085 inch thick for 30- and 24-inch diameter baskets, and not less than 0.0565 inch thick for 20-inch diameter baskets and shall be suitably perforated so as to freely discharge the water extracted. The side sheets shall be rigidly secured to a ferrous metal bottom which shall be covered with a plate of the same metal as the side, not less than 0.0450 inch thick for baskets 40 and 48 inches in diameter and not less than 0.0272 inch thick for smaller size baskets. The bottom of baskets 40 inches and over in diameter shall be formed to enclose the spindle and shall be taper bored to fit same. Baskets shall be mounted directly on the main spindle and shall be easily removable. The extractor drive shall be as specified in 3.5.2.4.

3.5.3.3.1 *Unloading type extractor containers.* Unless otherwise specified (see 6.2), each unloading type extractor shall be furnished with two sets of containers (4 sections). The sections shall be semicircular in shape, and the contour of the side sheets shall correspond exactly with the inner surface of the basket for the greater portion of its periphery in order to relieve the container from undue stress. Means shall be provided in the vertical flat sides to permit two containers to be hoisted up as one circular unit for depositing into, or lifting from the basket. Container bottoms shall be hinged at the back sheet and held in closed position by means of a self-locking po-

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sitive latch. The bottom shall be provided with wearing lugs which shall seat on the extractor basket bottom and serve as driving members. Three heavy-duty, swivel-type ball bearing casters shall be securely attached to each container section bottom to permit rolling of the containers over the floor. The container top ring shall be of corrosion-resisting metal not less than 0.0565 inch thick, or aluminum not less than minimum 0.105 inch thick, and shall be securely fastened to basket side sheets. Circular side sheets shall be fabricated from corrosion-resisting metal not less than 0.0565 inch thick and shall be perforated to permit free discharge of water during the extraction period. The vertical flat side sheet and bottom sheet of the containers shall be corrosion-resisting metal not less than 0.100-inch thick, unless reinforced galvanized steel construction is used, in which case the vertical side and bottom sheets shall be corrosion-resisting metal not less than 0.0335 inch thick, or aluminum not less than 0.105 inch thick. All seams of the container shall be tight and it shall be entirely smooth inside.

3.5.3.3.2 *Bottom covering plate.* The bottom covering plate, where used, shall extend between the side plate and bottom casting to the bottom of the side plate and shall be riveted or welded to the side plate bottom reinforcing band and the bottom of the basket. All seams in the plates shall be welded and made smooth by grinding and polishing. Spindle nut, basket-retaining member, and other parts, if inside of the basket, shall be of corrosion-resisting metal in accordance with 3.4.1.

3.5.3.3.3 *Basket top on rigid-curb and open-top extractors.* The top of the basket of the rigid-curb and open-top extractors shall be of the same material as the side. Unless the top of the basket is a casting or a plate not less than 0.177 inch thick for the 20- and 24-inch extractors, and not less than 0.240 inch thick for the 30-, 40-, and 48-inch extractors, it shall be reinforced by an endless ring welded or spun into place around the edge of the basket opening, or a spun or die-formed tubular section. Except as previously specified, the top shall be not less than 0.115 inch thick for the 40- and 48-inch sizes and not less than 0.071 inch thick for the 20-, 24-, and 30-inch sizes.

3.5.3.3.4 *Basket side reinforcing.* The basket side shall be reinforced, as required, by corrosion-resisting metal bands shrunk in place and of sufficient strength to prevent distortion of the basket when operating at its maximum capacity and speed. The bands shall be of a size not less than the manufacturer's standard design (see 4.4.2).

3.5.3.3.5 *Prevention of basket from striking the curb.* Means shall be provided to prevent the basket of the 20-, 24-, and 30-inch extractors from striking the curb in case of uneven loading (see 4.4.2).

3.5.3.4 *Spindle and spindle bearings.* The spindle shall be made of a type steel to insure safe operation, and shall be machined and ground to fit the supporting bearings. The spindle for extractors 40 inch and larger extractors shall be tapered and keywayed at the top to receive the basket hub and shall be threaded for accepting the top locking nut. If braking is applied on the spindle, the lower part of the spindle shall be keywayed to accept the spindle sheave and brakedrum.

3.5.3.4.1 *Keys and keyways.* Keys and keyways shall be of such design to insure alignment of the spindle and basket hub.

3.5.3.4.2 *Rigid-curb type extractor spindle bearing.* The spindle bearing for the rigid-curb type extractor shall be bronze with hardened steel convex steps located in the bottom, or ball- or roller-type bearings enclosed in a suitable housing in which the spindle shall rest. The bearing or bearing housing shall be flexibly mounted on or between heavy rings which shall be adjustable for compression.

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3.5.3.4.3 *Open-top type and unloading type extractors spindle bearings.* In open-top and unloading-type extractors the spindle shall rotate on ball or roller bearings in a housing.

3.5.3.5 *Signal lamp.* Extractors, 40 inch and larger, shall be provided with a signal lamp to indicate when the extractor is operating under power.

3.5.3.6 *Brake.* The brakedrum on the extractors shall be mounted on the lower end of the spindle or pulley or on the outside of the bottom casting of the basket. The brake shall be located so that the water being extracted does not come in contact with the brake lining. Brakes shall be accessible for relining.

3.5.3.6.1 *Footbrake.* Extractors up to and including 30 inch diameter shall be equipped with a footbrake having a shoe lined with renewable friction materials. The shoe shall be hinged and arranged to apply the proper pressure. Unless otherwise specified (see 6.2), the brake lever shall be nonlocking type. In lieu of the above type of footbrake, a selective timer and timer-controlled brake, as described in 3.5.3.6.2, may be used on all extractors up to and including 30 inches in diameter. Extractors shall be provided with a mechanical or electrical interlocking device which will prevent operation of the motor when either the brake is on or the safety cover is open, as described in 3.5.3.7 (see 4.4.2).

3.5.3.6.2 *Selective timer.* A selective timer shall be provided on 40 inch and larger extactors, in a location convenient to the operator, and arranged to stop the extractor basket at a predetermined time by automatically actuating an electrically, mechancially, hydraulically, or pneumatically operated or controlled brake capable of stopping the basket (see 4.4.2).

3.5.3.7 *Safety cover.* A perforated corrosion-resisting metal or aluminum safety cover shall be fitted and arranged so that it cannot be opened more than one inch while the basket is rotating more than 75 r.p.m. and the machine cannot be started when the cover is open more than one inch. This shall not prevent movement of the basket by hand, to permit even loading.

3.5.3.7.1 *Pneumatically operated cover.* A pneumatically-operated cover shall be supplied on the 54 and 60 inch size extractors, designed to operate at air pressure of from 70 to 75 pounds per square inch gage (p.s.i.g.) (see 4.4.2).

3.6 **Workmanship.** The completely fabricated extractor shall conform to the quality and grade of product covered by this specification. The extractor shall be free from defects such as fractures, splits, punctures, tears, dents, creases, bows, miscasts, deteriorations, or malformations. There shall be no sharp edges, burrs, or slivers.

3.6.1 *Application of finish.* The finish applied to the end item shall be continuous, smooth, adherent without discoloration or foreign material imbedded, and contain no sags, runs, drips, creeps, laps, bubbles, streaks, wrinkles, blisters, cracks, scratches, pores, pits, lumps, flux, or orange peel. No rust, rough grinds, or toolmarks shall show through the coating.

3.6.2 *Welders and welding.*

3.6.2.1 *Welders.* Before assigning any welder to equipment furnished under this specification, the contractor shall provide the applicable Government quality control office with the names of welders to be employed in the work, together with certificates that each of these welders has passed the qualification test as prescribed by either of the following listed codes for the type of welding operation to be performed and that such qualification is effective as defined by that particular code:

(a) Standard Qualification Procedures of the American Welding Society.
(b) Welding Qualification of the Ameri-

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can Society of Mechanical Engineers.

3.6.2.2 *Welding.* The surfaces of parts to be welded shall be free from oxide, scale, paint, grease, and other foreign matter. Welds shall be continuous, sound, smooth, and free from porosity, cracks, incomplete fusion, and deformation of material. All scale and flux (when flux is used) shall be removed from the finished welds.

3.6.3 *Soldering.* Soldering shall be complete and adherent with all flux and residue removed and shall contain no pinholes.

3.6.4 *Bolts, nuts, screws, studs and other threaded fasteners.* Threaded fasteners shall not be broken, stripped, fractured, or loose. All screw threads shall comply with the NBS Handbook H28. Unified threads shall be used in lieu of American National screw threads and nonunified American Standard screw threads.

3.6.5 *Driving of rivets.* Rivets shall be driven to completely fill the holes, with full size heads neatly finished and in full contact with the surface of the members.

3.6.6 *Electrical wiring.* Wiring shall not be cut, abraded, or have excessive insulation stripped, and shall be properly and tightly joined at terminals. Wiring shall have adequate slack to provide strain relief.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 *Army purchases.* For Army purchases, the testing performed by the contractor will be witnessed by an authorized representative of the Government prior to acceptance.

4.2 Preproduction sample inspection. When a preproduction sample is required, (see 3.1), it shall be examined for defects listed in table III, specified dimensions, and tested as specified in 4.4.1, 4.4.2, and 4.4.3.1 and when applicable 4.4.3.2.

4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated hereinafter.

4.3.1 *Component and material inspection.* In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.3.2 *End item inspection.* A lot shall consist of all extractors of the same type and size offered for inspection at one time. The sample unit shall be one completely fabricated extractor.

4.3.2.1 *Visual examination.* The completely assembled end item shall be examined for defects in finish, construction, workmanship, and marking. Defects shall be as classified in table III. The inspection level shall be level II with an acceptable quality level (AQL) of 2.5 for major defects and 6.5 for total defects, expressed in terms of defects per hundred units.

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TABLE III. *Classification of defects*

Examine	Defect	Classification	
		Major	Minor
Finish	Finish not as specified	X	
	Blistered, peeled, chipped, or area of *no film*		X
Materials	Materials not as specified	X	
Construction and workmanship, general (applicable to all components and assemblies)	Part missing	X	
	Fractured, split, punctured, dented, deteriorated	X	
	Not in proper alignment	X	
	Sharp burr or edge, sliver or splinter	X	
	Number of containers in type III extractors not as specified	X	
	Type III extractor baskets not hinged as specified and not provided with self-locking positive latch	X	
	Type III extractor containers not provided with type of casters specified	X	
	Spindle for 40 inch and larger extractors not tapered and keywayed as specified	X	
	Spindle bearing for type I extractors not of the type specified and not mounted as specified	X	
	Brakes not lined as specified and lining not readily accessible for relining	X	
	Brake lever for footbrake not of the nonlocking type (when applicable)	X	
	Conponent loose		X
	Not connected or joined as specified	X	
Construction and workmanship, general (applicable to all components and assemblies) (cont'd.)	Any adjustable assembly that cannot be adjusted to perform the function intended	X	
	Baskets not mounted on main spindle and not easily removable	X	
	Any component not readily accessible for servicing where required		X
	Drainage connections not of size specified and not in locations specified	X	
Welding and brazing (where required)	Missing, incomplete, burn holes, cracked, fractured, or otherwise not fused	X	
	Slag inclusion, slight undercut, not smooth and uniform, scale or *flux deposits not* removed		X
Soldering (when applicable)	Missing, not adherent or incomplete	X	
	Not clean (flux or flux residue not removed); not smooth (surface not finished neatly), or pinholes in solder		X
Bolts, nuts, screws, studs and other types of threaded fasteners	Missing, broken, stripped, fractured	X	
	Loose		X
Rivets (when applicable)	Broken, loose, not peened or insufficiently peened	X	
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminal, or not inclosed in specified conduit where required	X	

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TABLE III. *Classification of defects (cont'd)*

Examine	Defect	Classification	
		Major	Minor
Electrical assembly (cont'd.)	Adequate slack not provided for wiring to relieve strain or excessive insulation stripped from wiring------	X	
Lubrication (where applicable)	Not lubricated where required---	X	
	Lubrication fitting not accessible for servicing----------	X	
Marking for identification	Missing, incomplete, not legible-------		X
Instruction manual	Missing, incomplete, not legible---	X	

4.3.2.2 *Dimensional examination.* Inspection shall be made of the end item to determine compliance with dimensional requirements. Any dimension that is not within the specified requirements shall be classified a defect. The inspection level shall be S-2 with an AQL of 4.0 defects, expressed in terms of defects per hundred units.

4.3.2.3 *Testing the end item.* Unless otherwise specified (see 6.6), and when a preproduction sample is not required, the initial production unit shall be tested as specified in 4.4.2 and, when applicable, as specified in 4.4.3. Each production unit shall be tested as specified in 4.4.1. Any nonconformance shall be cause for rejection of the unit and the balance of the lot offered for inspection.

4.3.3 *Code and standards compliance.* Proof of compliance with the requirements of 3.3.1 and 3.3.2 shall be available to the Government representative. When moisture and fungus resistant treatment is specified the contractor shall furnish proof that electrical components comply with 3.5.2.12.

4.4 **Tests.**

4.4.1 *Operational testing.* Each extractor furnished under this specification shall be connected to power and operationally tested without garments or laundry. In the course of testing for operational requirements specified in 3.5.2, compliance with all performance requirements applicable to the fully assembled item, except those that require functional testing such as processing of garments or laundry through the equipment, shall be determined.

4.4.2 *Functional testing.* The extractor shall be tested by making the necessary connections to electric power, and compressed air, and by operating with saturated garments or laundry under full service load as specified in the applicable paragraphs of this specification.

The following items and operations shall be checked as applicable:

(a) Leakage in curb, overheating of moving parts and bearings, operation of starting and stopping devices, and structural failures for compliance with the requirements of 3.5.2.

(b) Starting of motors, without overheating, for compliance with the requirements of 3.5.2.1.2.2.

(c) Anchoring of the machine and proper contact with the floor for compliance with the requirements of 3.5.2.2.

(d) Lubrication for compliance with the requirements of 3.5.2.7.

(e) Basket speed for compliance with the requirements of 3.5.3.1 and table I.

(f) Construction of top of curb for ready removal of basket and means for prevention of splashing of water, for compliance with the requirements of 3.5.3.2.1.

(g) Strength of basket side reinforcements for compliance with the requirements of 3.5.3.3.4.

(h) Prevention of baskets of 20-, 24-, and 30-inch size extractors from striking the curb for compliance with the requirements of 3.5.3.3.5.

(i) Brake for 30-inch and smaller size

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extractors for compliance with the requirements of 3.5.3.6.1.

(j) Selective timer provided on 40 inch and larger size extractors for compliance with the requirements of 3.5.3.6.2.

(k) Cover of 54- and 60-inch size extractors for compliance with the requirements of 3.5.3.7.1.

4.4.3 *Radio interference suppression test.*

4.4.3.1 *Army purchases.* The preproduction sample or initial unit of production equipped for the reduction of radio interference in accordance with 3.5.2.8.1, shall be tested as described in MIL-E-55301 by the contractor or his private testing organization. The Government reserves the right to witness tests performed by the contractor or his private testing organization. The contractor shall furnish the contracting officer, for approval, the electromagnetic compatibility test report required in MIL-E-55301. The report(s) shall include all the required information and shall be submitted within the time interval specified in MIL-E-55301. Upon approval of the report(s) by the contracting officer and provided all other requirements of the specification are met, the preproduction sample or initial unit of production shall be used as a model for all production units. If the contractor can certify that the equipment utilizes induction squirrel-cage type motors and the switching transients can be exempted as short duration interference (see 3.4.1 of MIL-E-55301) the contracting officer may accept the units without tests or a report.

4.4.3.2 *Navy and Air Force purchases.* When required, sampling, inspection, and test procedures shall conform to the requirements of MIL-I-16910 (see 3.5.2.8.2). Tests for radiated and conducted interference on the complete equipment assembly shall be conducted on the production test unit and upon such additional units as are specified. For tests on the complete equipment assembly, all testing equipment, instruments, personnel for making the tests, the test location which shall be reasonably free from radiated and conducted interference, and other necessary facilities shall be furnished by the contractor.

4.5 Examination of preparation for delivery. An examination shall be made to determine that packaging, packing, and marking requirements of section 5 are complied with. The examination shall be in accordance with the quality assurance provisions of MIL-L-3153 except that the inspection level shall be S-2 with an AQL of 4.0 defects per hundred units.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Preservation, packaging, and packing shall be in accordance with the applicable level A, B, or C requirements of MIL-L-3153, as specified (see 6.2).

5.2 Marking.

5.2.1 *Civil agencies.* In addition to any special marking required by the contract or order (see 6.2), interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 *Military agencies.* In addition to any special marking required by the contract or order (see 6.2), interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The laundry extractors covered by this specification are intended for use in fixed installations.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(*a*) Title, number, and date of this specification.

(*b*) Type and size required (see 1.2.1).

(*c*) When preproduction sample is required (see 3.1).

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(*d*) Electrical characteristics (voltage, phase, and frequency) (see 3.5.2.1.1).
(*e*) Speed warning on basket larger than 24-inches, when required (see 3.5.2.5).
(*f*) When radio interference suppression is required for purchases for Air Force and Navy (see 3.5.2.8.2).
(*g*) Finish and color, when other than specified is required (see 3.5.2.9).
(*h*) Quantity of instruction books required if different than specified (see 3.5.2.10.1).
(*i*) Quantity of repair parts manuals required if different than specified (see 3.5.2.10.2).
(*j*) When moisture and fungus-resistant treatment is required (see 3.5.2.12).
(*k*) Number of containers required with each type III extractor if other than specified (see 3.5.3.3.1).
(*l*) When locking-type lever is required for foot-operated brake on type I extractors (see 3.5.3.6.1).
(*m*) Selection of applicable levels of preservation, packaging, and packing (see 5.1).
(*n*) Any special marking for shipping required (see 5.2).

6.3 Drawings and services. When drawings are required for installation purposes, the contracting officer should request the contractor to furnish such drawings showing the exact dimensions of the extractor, overall floor space requirement, location of holddown bolts, location of the inlet and outlet lines, the point where electrical connection is to be made, and the electrical characteristics of the motor and other electrical equipment. When services are required by the contracting officer, he should request the contractor to furnish supervision of installation and demonstration services of the extractor. When such services are required, the location, approximate date, the person or office to report to, and any accommodations or services to be supplied by the installation or agency should be stated.

6.4 Radio interference reduction tests (Army purchases). The contracting officer should submit a copy of the report(s), required by 4.4.3.1, to the Commanding General, U.S. Army Electronics Command, ATTN: AMSEL-RD-GF, Fort Monmouth, New Jersey for approval.

6.5 Successful commercial operation. Invitation for bids should specify that no item of equipment will be acceptable unless the manufacturer has had equipment of approximately the same design as that specified operating successfully in a commercial or institutional laundry for at least 1 year. Equipment installed for test purposes in a manufacturer's plant or laboratory shall not come within the category of successful commercial operation.

6.6 Functional testing. The functional tests required by 4.3.2.3 for the initial production unit may be waived by the contracting officer if these tests have been performed previously on an identical extractor and the contractor provides certified test data indicating compliance with the requirements.

6.7 Repair parts for overseas shipments. For overseas shipments, the contracting officer should make certain that one year's supply of high mortality repair parts are included with the equipment.

6.8 Level B packaging for civil agency procurements. When level B packaging is required for civil agency procurements the level A packaging requirements of MIL-L-3153 shall apply.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-D-26549 (USAF)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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